ECHO: The Signal That Was Never Sent

Case Report · Entangled Minds: The Quantathon

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Case file: QX-ECHO-2026
Modules: Mandatory 01, 03, 05 · Stretch 06

1. Final Theory and Verdict

ECHO did not send a message faster than light. My theory is that Dr. Zara Mahmud saw genuine quantum correlations between her two detectors and took their perfect agreement for a signal. Entanglement makes two outcomes match, but each outcome is random, so neither side can choose what the other sees.

Confidence: high (about 95%) that ECHO was not faster-than-light. Moderate (about 60%) that her detectors were really entangled and she only misread them (see section 6). Low for anything about her disappearance or who is responsible.

2. Evidence and Clues

1. **Agreement is not proof (Module 03).** Two people sharing one hidden bit also agree every time: 980 × 00 and 1,020 × 11 in my classical baseline, against 992 and 1,008 for the Bell pair. So Zara's perfect agreement could not prove a link, let alone a message.
2. **CHSH (Module 03).** Measuring at different angles gave S = 2.842 ± 0.022. No pre-agreed classical strategy can exceed 2, so the Bell-pair correlations are real in the simulation, but they still carry no message.
3. **Teleportation (Module 06).** Without Alice's two classical bits, Bob saw 974 zeros and 1,026 ones, a coin flip. With them, all 2,000 shots gave 0. The state only arrives once ordinary classical information reaches Bob.

3. Evidence Board

FACT	The 96 supplied bits decode to ECHO (05). H–H gives 1,000/1,000 zeros (01). Bell pair gives only 00 and 11 (03). CHSH S = 2.842 ± 0.022 (03). Without corrections Bob sees 974/1,026 (06).
CLAIM	Zara: ECHO was faster than light. Vance: "ECHO was never real." Rao: "I only deleted a corrupted log." Bilal: "I warned her" (his report is missing). ARIA: "No unauthorized command was issued." I cannot verify these.
INFERENCE	My results fit entanglement giving correlation only. ECHO fits the machine looping its own project name. Rao's deletion means the log timestamps cannot be fully trusted.
OPEN QUESTION	Who configured the final run? What did the deleted log contain? What was in Bilal's report? What happened to Zara? Were her real detectors entangled at all?

4. Experiments and Results

All runs use Qiskit Aer (simulation). For each, I state the expected result, then the observed one.

- **Module 01.** Expected: X → 1, H → about 50/50, H–H → 0, X on qubits 0 and 2 → 101. Result: 1,000 ones; 517 zeros / 483 ones; 1,000 zeros; 101 in 1,000 shots. "Looks random" is not the same as "is random" (H–H returns to 0 by interference).
- **Module 03.** Expected: a Bell pair (H then CNOT) gives only 00 and 11 in both Z and X. Result: 992/1,008 (Z), 995/1,005 (X). Since a hidden shared bit also matches in Z, I ran CHSH (angles 0, $\pi/2$ and $\pi/4$, $-\pi/4$): $S = 2.842 \pm 0.022$. The ideal limit is 2.828, so my value is about 0.6 standard errors above it, which is shot noise.
- **Module 05.** Majority voting on the 32 triples gives 01000101 01000011 01001000 01001111, which reads ECHO. Two flips in one triple (000 → 110) decode wrongly with no warning, so the code corrects one flip per triple.
- **Module 06.** Teleporting $|+\rangle$ with classical feed-forward gave 2,000/2,000 zeros. Without the corrections: 974/1,026.

5. Interpretation

The crux is agree versus send. The Bell pair makes the detectors agree, but each result is a fair coin, so Alice cannot choose her result to encode a bit. Bob only discovers the agreement when they compare notes over an ordinary channel. Teleportation shows the same thing: Bob needs Alice's classical bits. These results challenge Zara's claim, but cannot prove how the real hardware or logs behaved, or who ran the final experiment.

6. Alternative Explanation

Zara's detectors were never entangled. Their agreement was classical or arranged, for example a pre-agreed record or a configuration set through ARIA by someone with access (Rao or Vance). Module 03 shows agreement in one basis can be produced exactly this way, and I did not test her real detectors. I prefer my theory because it needs no hidden actor and fits the simplest reading, but my confidence on this point is only moderate.

7. Next Experiment

Run a CHSH-style test on Zara's actual detector terminals, with each side choosing its angle at random. If my theory is right, S should be near 2.8. If the rival is right (classical or arranged agreement), S should not exceed 2. Same-basis agreement looks identical under both, so only different-angle data separates them.

8. Limitations

Everything is an ideal, noiseless simulation. CHSH in a simulator shows what quantum mechanics allows, not what Zara's detectors did. Teleportation was tested only with $|+\rangle$, and I did not use the original seed circuit, so I make no claim about Module 02. None of this shows where Zara is or who is responsible.

References and AI Assistance

1. ECHO Participant Manual and Code of Conduct (Qubits Qrew, NED University).
2. Clauser, Horne, Shimony, Holt (1969), Phys. Rev. Lett. 23, 880.
3. Wootters and Zurek (1982), Nature 299, 802.
4. Qiskit documentation, https://docs.quantum.ibm.com/

I used ChatGPT and Claude (Anthropic) to review the notebook, check my reasoning and organise this report. The circuits were run in Google Colab with Qiskit Aer, and every number above comes from the notebook's saved outputs.

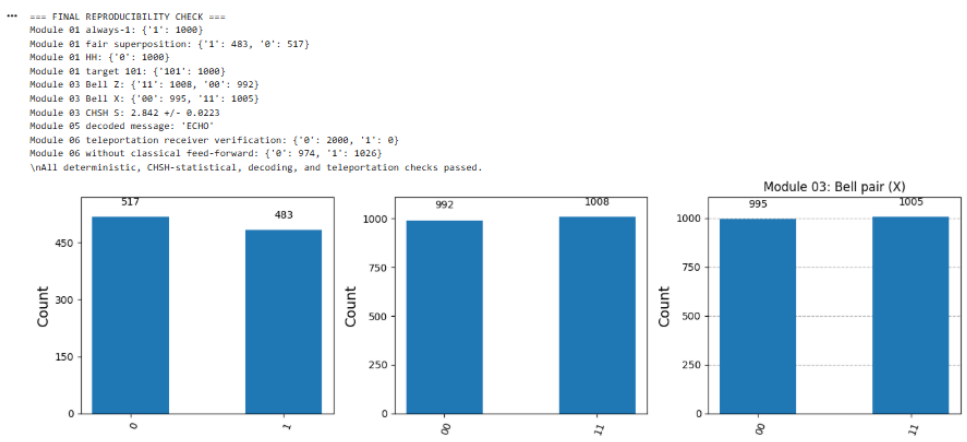


Figure 1:Notebook final-check output: counts, decoded word ECHO, CHSH S = 2.842 ± 0.0223, and histograms.

Conclusion

The results do not support faster-than-light communication. The Bell pair gave matching outcomes and CHSH gave S = 2.842, so the correlations are real in the simulation, but each outcome is random and Bob's qubit stayed a coin flip (974/1,026) until Alice's classical bits were used. They cannot show whether Zara's real detectors were entangled, or what happened to her.